

Class 7 More Proofs

Goals: Be able to prove implication statements, p implies q , by contrapositive and by contradiction.

Last class we learned how to prove an existential statement by giving an example. We learned how to disprove a universal statement by giving a counterexample and we practiced proving an implication by giving a direct proof.

Today we learn two more techniques to prove that an implication is true.

1. Proof by contrapositive

Our strategy for a proof of $p \rightarrow q$ by contrapositive is to assume $\sim q$ is given and show that $\sim p$ follows.

Example: Prove if the product xy is even then either x or y is even.

We will do this using a direct proof of the contrapositive statement.

So we want to prove if x and y are odd then xy is odd.

2. Proof by Contradiction

Our strategy for a proof of $p \rightarrow q$ by contradiction is to assume p and $\sim q$ are given then show this leads to a contradiction.

Example: Prove that $\sqrt{2}$ is irrational.

Rational or Irrational Proofs

Write the contrapositive of each statement in English. Use the word irrational to indicate not rational.

Prove or disprove each statement below. Describe which method of proof (or disproof) used.

1. If both a and b are rational numbers then $a+b$ is rational.
2. If a is rational and b is irrational then $a+b$ is irrational.
3. If both a and b are irrational then $a+b$ is irrational.
4. If $a+b$ is rational then both a and b are rational.
5. If $a+b$ is irrational then at least one of a or b must be irrational.